



Inspiring Excellence

Assignment-02

Karnaugh Map (K-MAP)

Name: Md. Tahsin Hossain

ID: 24301099

Course: CSE260

Section: 17B

Submitted to: Majisha Jahan Disha [MAJI]

a

Min Terms	P	Q	R	S	A	W
0	0	0	0	0	1	X
1	0	0	0	1	X	0
2	0	0	1	0	1	X
3	0	0	1	1	X	0
4	0	1	0	0	0	0
5	0	1	0	1	X	0
6	0	1	1	0	0	1
7	0	1	1	1	0	0
8	1	0	0	0	1	X
9	1	0	0	1	0	1
10	1	0	1	0	0	0
11	1	0	1	1	0	0
12	1	1	0	0	0	1
13	1	1	0	1	X	0
14	1	1	1	0	0	1
15	1	1	1	1	0	0

b

PQ \ RS	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$A = \bar{P}\bar{Q} + \bar{Q}\bar{R}\bar{S}$$

PQ \ RS	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$W = P\bar{Q}\bar{R} + QR\bar{S} + PQ\bar{S}$$

ASSIGNMENT 2: The CSE2b team is building a system for a spacecraft that aims to detect two specific signals: **Signal Alpha** and **Signal Omega**. The spacecraft is equipped with four types of pulse detection sensors: **P, Q, R, and S**. The system processes the pulse values from these sensors and detects **Signal Alpha** and **Signal Omega** based on the **pulse patterns**.

Inputs:

The system has four pulse detection sensors (P, Q, R, S). For each sensor:

- **Input = 1** if a pulse is detected.
- **Input = 0** if no pulse is detected.

Outputs:

- **Signal Alpha**
- **Signal Omega**

Conditions for Detecting Signal Alpha:

1. If the decimal value of the pulse sequence of the four sensors (**P, Q, R, S**) results in an **even number** in the **Fibonacci series**, then **Signal Alpha** is detected.
 - **Recall:** The **Fibonacci series** starts as: **0, 1, 1, 2, 3, ...** (i.e., each number is the sum of the previous two numbers).
 - The **even Fibonacci numbers** in the series are: **0, 2, ...**
2. If the decimal value of the pulse sequence of the four sensors results in an **odd number** in the Fibonacci series then the spacecraft's detection module **cannot correctly identify Signal Alpha**. In this case, the output for **Signal Alpha** is **Don't Care**.
3. For all other combinations of pulses, **Signal Alpha** is **not detected**.

Conditions for Detecting Signal Omega:

1. If there are **consecutive 1s** in the pulse sequence (e.g., $P = 1, Q = 1, R = 0, S = 0$), and the resulting number is **not odd** (Here 1100 equals 12 which is even), then **Signal Omega** is detected.
 - The number of **1s** in the sequence must be **greater than 1** to consider it as consecutive 1s (e.g., $P = 1, Q = 0, R = 0, S = 0$ does not have consecutive 1s).
2. If there are **consecutive 0s** (number of 0s greater than 1) in the pulse sequence, and **Pulse P** is **detected** then **Signal Omega** is detected.
3. **Signal Omega** cannot be detected if **Signal Alpha** has already been detected for the same combination of pulses. In this case, the output for **Signal Omega** is **Don't Care** (whether the

0, 1, 1, 2, 3, 5, 8, 13,

conditions for Signal Omega are met or not).

4. For all other combinations of pulses that do not meet the above criteria, **Signal Omega is not detected.**

Help the CSE2b team establish the system:

- a) Create a truth table for the system described above. For the outputs, use the variables **A** and **W**, and use **Don't Care** values where necessary. For the inputs, maintain the order as **P, Q, R, S**, where **P** is the MSB and **S** is the LSB. [3]
- b) Use Karnaugh map(s) to find the simplified expressions for each of the outputs. [2]

DEADLINE : 22 / 07 / 2026 at 11 59 pm

In case of late Submissions, 10% marks will be deducted for every 12 hours of delay.

Submission Link: <https://forms.gle/vyqUmKNNfGv3QZCA8>